Power Series and Interval of Convergence

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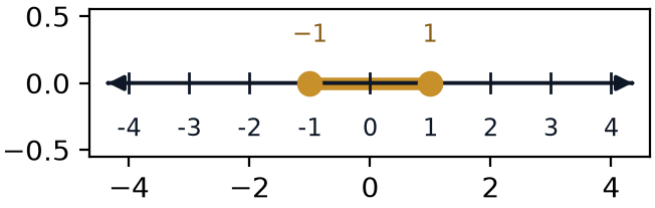
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Finding the radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.

2 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.



3 [Foundational] • Target: Representing functions as power series

Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.

4 [Foundational] • Target: Representing functions as power series

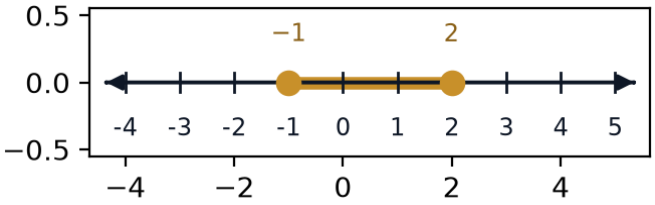
Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.

5 [Foundational] • Target: Differentiating and integrating power series

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.

6 [Foundational] • Target: Differentiating and integrating power series

Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.



7 [Foundational] • Target: More radius/interval of convergence practice

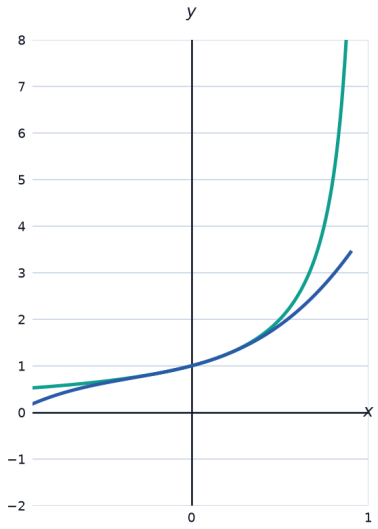
Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n!\, x^n$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Visualizing a power series approximation

The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.

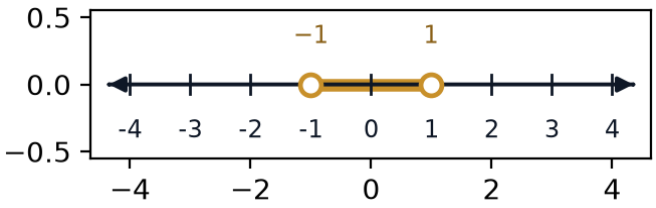


9 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.

10 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.



11 [Intermediate] • Target: Building a new series by combining known ones

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

12 [Intermediate] • Target: Ratio Test with factorial and power combinations

For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

13 [Intermediate] • Target: Using a power series to approximate a definite integral

Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

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14 [Intermediate] • Target: Differentiating and integrating power series

Using the binomial series for $(1-x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x)=\arcsin x$ up to and including the x^5 term. State its radius of convergence.

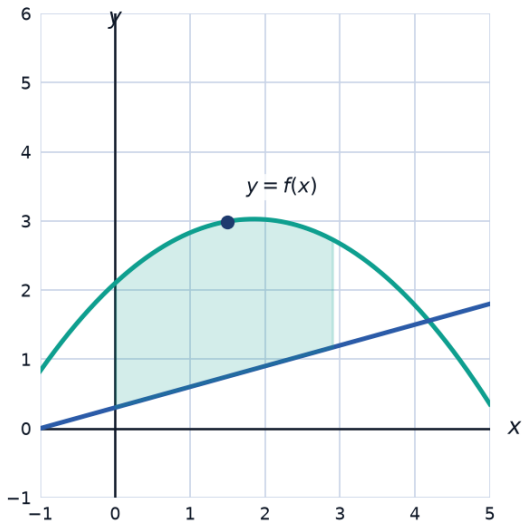
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 3.01 Parametric Equations

Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

16 [Challenging] • Target: Visualizing a power series approximation • Cross-Lesson: 3.02 Vector-Valued Functions

Level 3 · AdvancedEvaluate the definite integral using a substitution that transforms the limits: $\int_0^{\sqrt{\pi/2}} x\cos(x^2)e^{\sin(x^2)}\,dx$



17 [Challenging] • Target: Term-by-term operations preserve the radius of convergence • Cross-Lesson: 4.01 Polar Curves and Derivatives

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0,1]$ and evaluate it: $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{i^3}{n^4}$.

18 [Challenging] • Target: Building a new series by combining known ones • Cross-Lesson: 4.02 Area in Polar Coordinates

Use the substitution $x=2\sin\theta$ to evaluate the exact definite integral $\int_0^1\sqrt{4-x^2}\,dx$.

19 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3}\left(x^5\sqrt{9-x^2}+\sqrt{9-x^2}\right)dx$.

20 [Challenging] • Target: Using a power series to approximate a definite integral • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int\frac{5x-3}{x^2-2x-3}\,dx$

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding the radius of convergence

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x+7)^n}{3^n}$.

a. [-10,-4]

b. [-10,-4)

c. (-10,-4)

d. (-10,-4]

22 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^3.

a. 27

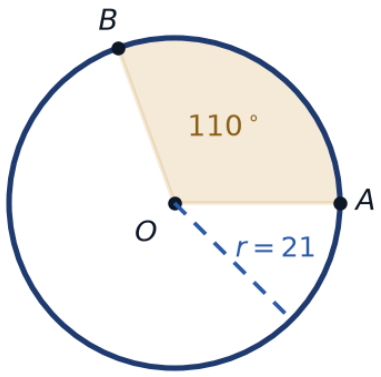
b. 26

c. 81

d. 9

23 [Foundational] • Target: Representing functions as power series

Integrating $\frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k}$ term-by-term gives the Maclaurin series for $\arctan x$. Find the coefficient of x^{21}.



a. $\frac{2}{21}$

b. $-\frac{1}{21}$

c. $\frac{1}{21}$

d. $\frac{1}{22}$

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24 [Foundational] • Target: Differentiating and integrating power series

Differentiating the Maclaurin series $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ term-by-term gives the series for $\frac{1}{(1-x)^2}$. Find the coefficient of x^3.

a. 6

b. 8

c. 4

d. 3

25 [Foundational] • Target: More radius/interval of convergence practice

Interval of convergence of $\sum \frac{(-3)^n (x-2)^n}{\sqrt{n}}$: centre, radius, and endpoints:

a. $R = 1/3$, interval $(5/3, 7/3]$

b. $R = 1/3$, interval $[5/3, 7/3)$

c. $R = 1/3$, interval $[5/3, 7/3]$

d. $R = 3$, interval $(-1, 5]$

26 [Foundational] • Target: Visualizing a power series approximation

The function $f(x)$ is represented by the power series $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$. It is known that $f(0) = 2$, $f'(0) = 3$, and $f''(0) = 12$. If the Taylor series expansion of $f(x)$ around $x = 0$ is given by $f(x) = 2 + 3x + bx^2 + \ldots$, find the value of the coefficient b and the center c.

a. $b = 6, c = 0$

b. $b = 6, c = 1$

c. $b = 12, c = 0$

d. $b = 2, c = 0$

27 [Foundational] • Target: Term-by-term operations preserve the radius of convergence

Consider the Taylor series expansion of the function $f(x) = e^{kx}$ centered at $x = 0$. If the coefficient of the x^3 term in this expansion is $\frac{4}{3}$, what is the value of the constant k?

a. $k = 1$

b. $k = 2$

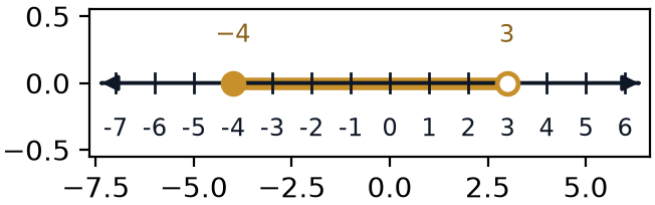
c. $k = 3$

d. $k = 4$

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28 [Foundational] • Target: Building a new series by combining known ones

Consider the function $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. What is the value of the coefficient of the x^3 term in the Maclaurin series for $f(x)$?



a. $\frac{1}{3!}$

b. $\frac{1}{3}$

c. $\frac{1}{6}$

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Ratio Test with factorial and power combinations

In the Maclaurin series of $\sin(6x)$, what is the coefficient of the x^2 term (even power)?

a. 18

b. -18

c. 1

d. 0

30 [Intermediate] • Target: Using a power series to approximate a definite integral

Consider the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 3^n}$. This series defines a function $f(x)$ over its interval of convergence. Let I be the interval of convergence for this power series. Determine which of the following statements is TRUE about $f(x)$ and its behavior at the endpoints of I.

a. The radius of convergence is 3, and the series converges at both endpoints of the interval of convergence.

b. The radius of convergence is 3, and the series diverges at the left endpoint and converges at the right endpoint.

c. The radius of convergence is $1/3$, and the series converges at both endpoints of the interval of convergence.

d. The radius of convergence is $1/3$, and the series diverges at both endpoints of the interval of convergence.

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31 [Intermediate] • Target: Finding the radius of convergence

Term-by-term differentiation is valid inside the radius of convergence. The derivative of $\sum_{n=0}^{\infty} x^n/(n+1)$ is:

a. $\sum x^n/n$

b. $\sum nx^{n-1}/(n+1)$

c. $1/(1-x)$

d. $\ln(1-x)$

32 [Intermediate] • Target: Finding the interval of convergence (testing endpoints)

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R=3$ and the center of the series is $a=2$, which of the following intervals represents the values of x for which the series converges?



a. $(-1, 5)$

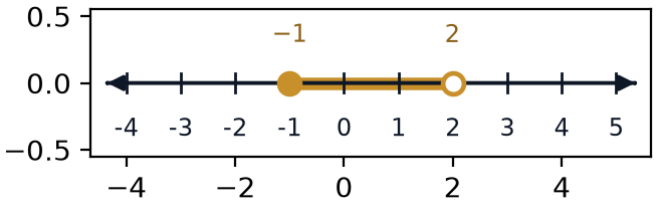
b. $[-1, 5]$

c. $(-1, 5]$

d. $[-1, 5)$

33 [Intermediate] • Target: Representing functions as power series

Consider the function $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$. Which of the following is the Maclaurin series for $\cos(x)$?



a. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

b. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

c. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

d. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$

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34 **[Intermediate]** • **Target: Differentiating and integrating power series**

What is the Maclaurin series expansion for $f(x) = e^{3x}$?

a. $\sum_{n=1}^{\infty} \frac{(3x)^n}{n}$

b. $\sum_{n=0}^{\infty} (-1)^n \frac{(3x)^n}{n!}$

c. $\sum_{n=0}^{\infty} \frac{3x^n}{n!}$

d. $\sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$

35 **[Intermediate]** • **Target: More radius/interval of convergence practice**

Find the coefficient of x^4 in the Maclaurin series of e^{2x}.

a. $\frac{2}{3}$

b. $\frac{4}{3}$

c. $\frac{8}{3}$

d. $\frac{1}{12}$

36 **[Intermediate]** • **Target: Visualizing a power series approximation**

Given $f^{(3)}(a) = 4$, find the coefficient of $(x - a)^3$ in the Taylor series of f centred at a.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. 2

d. 4

37 **[Intermediate]** • **Target: Term-by-term operations preserve the radius of convergence**

Maclaurin series for $\sin x$ is:

a. $\sum \frac{x^{2n}}{(2n)!}$

b. $\sum (-1)^n \frac{x^{2n+1}}{(2n+1)!}$

c. $\sum (-1)^n \frac{x^{2n}}{(2n)!}$

d. $\sum \frac{x^n}{n!}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Building a new series by combining known ones • Cross-Lesson: 4.02 Area in Polar Coordinates

Taylor series for e^x at $x = 0$ is:

a. $\sum \frac{x^n}{n}$

b. $\sum \frac{x^n}{n!}$

c. $\sum x^n$

d. $\sum (-1)^n \frac{x^n}{n!}$

39 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Composite series: $e^{\sin x}$ up to x^3. Using $\sin x = x - x^3/6 + \cdots$ inside $e^u = 1 + u + u^2/2 + \cdots$:

a. $1 + x + \frac{x^2}{2}$

b. e

c. $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

d. $1 + x + \frac{x^2}{2} - \frac{x^3}{3}$

40 [Challenging] • Target: Using a power series to approximate a definite integral • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For a power series $\sum c_n x^n$, the radius of convergence R satisfies $1/R =$

a. $\limsup |c_n|^{1/n}$

b. $\limsup |c_{n+1}/c_n|$

c. $\sup |c_n|$

d. $|c_0|$

41 [Challenging] • Target: Finding the radius of convergence • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Using the power series $1/(1 - x) = \sum x^n$, find $\sum n x^{n-1} =$

a. $1/(1 - x)$

b. $1/(1 - x)^2$

c. $x/(1 - x)^2$

d. $1/(1 + x)^2$

42 [Challenging] • Target: Finding the interval of convergence (testing endpoints) • Cross-Lesson: 1.01 Integration by Parts

Radius of convergence of $\sum x^n/n$ is $R =$

a. 0

b. e

c. ∞

d. 1

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43 [Challenging] • Target: Representing functions as power series • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A power series $\sum c_n x^n$ with radius of convergence R converges absolutely for:

a. no x

b. all x

c. |x|<R

d. $|x| > R$

44 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find radius of convergence: $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$:

a. 1/4

b. 1/2

c. ∞

d. 4

45 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 3.01 Parametric Equations

If the radius of convergence of $\sum a_n x^n$ is R, then the radius of $\sum a_n x^{2n}$:

a. R

b. R^2

c. $\sqrt{R}$

d. $2R$

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